

# Row Echelon Form — 20 Solved Examples

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## Problem 1 (B.S. Grewal)

Matrix:

$$A = \begin{pmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{pmatrix}$$

Task: Reduce to Row Echelon Form and find rank.

Solution (step-by-step):

1. Start with  $A = \begin{pmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{pmatrix}$ .

2. Use  $R_2 \leftarrow R_2 - 2R_1$ :

$$R_2 = (4, 9, -3) - 2(2, 4, -2) = (4 - 4, 9 - 8, -3 + 4) = (0, 1, 1).$$

Matrix becomes  $\begin{pmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ -2 & -3 & 7 \end{pmatrix}$ .

3. Use  $R_3 \leftarrow R_3 + R_1$ :

$$R_3 = (-2, -3, 7) + (2, 4, -2) = (0, 1, 5).$$

Matrix  $\begin{pmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ 0 & 1 & 5 \end{pmatrix}$ .

4. Use  $R_3 \leftarrow R_3 - R_2$ :

$$R_3 = (0, 1, 5) - (0, 1, 1) = (0, 0, 4).$$

Matrix  $\begin{pmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 4 \end{pmatrix}$ . — This is in row echelon form (leading entries 2,1,4 down the diagonal).

5. All three rows are nonzero so **rank** = 3.

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## Problem 2 (B.S. Grewal)

Matrix:

$$B = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{pmatrix}$$

Task: REF and rank.

Solution:

1. Start:  $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{pmatrix}$ .

2.  $R_2 \leftarrow R_2 - 2R_1 \rightarrow R_2 = (0, 0, 0)$ . Matrix becomes  $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$ .

3.  $R_3 \leftarrow R_3 - R_1 \rightarrow R_3 = (0, -1, -2)$ . Matrix  $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{pmatrix}$ .

4. Swap  $R_2$  and  $R_3$  to get nonzero row up:  $\begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$ . — REF achieved.

5. Two nonzero rows  $\rightarrow$  **rank = 2**.

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### Problem 3 (B.S. Grewal)

Matrix:

$$C = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 4 & 5 \\ 3 & 5 & 7 & 9 \end{pmatrix}$$

Task: REF and rank.

Solution:

1. Start:  $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 4 & 5 \\ 3 & 5 & 7 & 9 \end{pmatrix}$ .

2.  $R_2 \leftarrow R_2 - 2R_1 \rightarrow R_2 = (0, 1, 2, 3)$ .

3.  $R_3 \leftarrow R_3 - 3R_1 \rightarrow R_3 = (0, 2, 4, 6)$ . Matrix now  $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 2 & 4 & 6 \end{pmatrix}$ .

4.  $R_3 \leftarrow R_3 - 2R_2 \rightarrow R_3 = (0, 0, 0, 0)$ . So REF is  $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ .

5. Two nonzero rows  $\rightarrow$  **rank = 2**.

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### Problem 4 (B.S. Grewal)

Matrix:

$$D = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 3 & 4 \\ 2 & 5 & 8 \end{pmatrix}$$

Task: REF and rank.

Solution:

1. Swap  $R_1 \leftrightarrow R_2$  to get a nonzero pivot first:  $\begin{pmatrix} 1 & 3 & 4 \\ 0 & 1 & 2 \\ 2 & 5 & 8 \end{pmatrix}$ .
  2.  $R_3 \leftarrow R_3 - 2R_1$ :  $R_3 = (2, 5, 8) - 2(1, 3, 4) = (0, -1, 0)$ . Matrix  $\begin{pmatrix} 1 & 3 & 4 \\ 0 & 1 & 2 \\ 0 & -1 & 0 \end{pmatrix}$ .
  3.  $R_3 \leftarrow R_3 + R_2$ :  $R_3 = (0, -1, 0) + (0, 1, 2) = (0, 0, 2)$ . REF:  $\begin{pmatrix} 1 & 3 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{pmatrix}$ .
  4. All rows nonzero  $\rightarrow$  rank = 3.
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### Problem 5 (B.S. Grewal)

Matrix:

$$E = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 6 & 1 & 4 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

Task: REF and rank.

Solution:

1. Start with  $R_1$  as pivot row:  $\begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 6 & 1 & 4 \\ 0 & 0 & 1 & 1 \end{pmatrix}$ .
2.  $R_2 \leftarrow R_2 - 2R_1 \rightarrow R_2 = (0, 0, 1, 1)$ .  
 $R_3 \leftarrow R_3 - 3R_1 \rightarrow R_3 = (0, 0, 1, 1)$ . Matrix  $\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$ .

3. Now pivot at column 3:  $R_3 \leftarrow R_3 - R_2 = (0, 0, 0, 0)$ .  $R_4 \leftarrow R_4 - R_2 = (0, 0, 0, 0)$ . Matrix

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

4. Two nonzero rows  $\rightarrow$  **rank** = 2.

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### Problem 6 (B.S. Grewal)

Matrix:

$$F = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 2 & -1 & 3 \end{pmatrix}$$

Task: REF and rank.

Solution:

1. Use  $R_3 \leftarrow R_3 - 2R_1$ :  $R_3 = (2, -1, 3) - 2(1, 0, 2) = (0, -1, -1)$ . Matrix  $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & -1 & -1 \end{pmatrix}$ .

2.  $R_3 \leftarrow R_3 + R_2$ :  $(0, -1, -1) + (0, 1, -1) = (0, 0, -2)$ . REF:  $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & -2 \end{pmatrix}$ .

3. Three nonzero rows  $\rightarrow$  **rank** = 3.

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### Problem 7 (B.S. Grewal)

Matrix:

$$G = \begin{pmatrix} 2 & 4 & 1 \\ 1 & 2 & 0 \\ 3 & 6 & 1 \end{pmatrix}$$

Task: REF and rank.

Solution:

1. Swap  $R_1 \leftrightarrow R_2$  (makes simpler pivot):  $\begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 3 & 6 & 1 \end{pmatrix}$ .

2.  $R_2 \leftarrow R_2 - 2R_1 \rightarrow R_2 = (0, 0, 1)$ .

$R_3 \leftarrow R_3 - 3R_1 \rightarrow R_3 = (0, 0, 1)$ . Matrix  $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ .

3.  $R_3 \leftarrow R_3 - R_2 \rightarrow R_3 = (0, 0, 0)$ . So REF:  $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ .

4. Two nonzero rows  $\rightarrow$  **rank** = 2.

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### Problem 8 (B.S. Grewal)

Matrix:

$$H = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 1 & 3 & 5 & 7 \end{pmatrix}$$

Task: REF and rank.

Solution:

1.  $R_3 \leftarrow R_3 - R_1: R_3 = (0, 1, 2, 3)$ . Now matrix  $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{pmatrix}$ .

2.  $R_3 \leftarrow R_3 - R_2: R_3 = (0, 0, 0, 0)$ . REF:  $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ .

3. Two nonzero rows  $\rightarrow$  **rank** = 2.

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### Problem 9 (B.S. Grewal)

Matrix:

$$K = \begin{pmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \\ 1 & 0 & 1 \end{pmatrix}$$

Task: REF and rank.

Solution:

1.  $R_2 \leftarrow R_2 - 2R_1 \rightarrow R_2 = (0, 0, 0)$ . Matrix  $\begin{pmatrix} 1 & -1 & 2 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ .

2.  $R_3 \leftarrow R_3 - R_1 \rightarrow R_3 = (0, 1, -1)$ . Swap to get it in row 2:  $\begin{pmatrix} 1 & -1 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$ .

3. Two nonzero rows  $\rightarrow$  **rank** = 2.

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### Problem 10 (B.S. Grewal)

Matrix:

$$L = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Task: REF and rank.

Solution:

1.  $R_2 \leftarrow R_2 - 4R_1 \rightarrow R_2 = (0, -3, -6).$

$R_3 \leftarrow R_3 - 7R_1 \rightarrow R_3 = (0, -6, -12).$  Matrix  $\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{pmatrix}.$

2.  $R_3 \leftarrow R_3 - 2R_2 \rightarrow R_3 = (0, 0, 0).$  REF:  $\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{pmatrix}.$

3. Two nonzero rows  $\rightarrow$  **rank = 2.**

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### Problem 11 (R.K. Jain & Iyengar)

Matrix:

$$M = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

Task: REF and rank.

Solution:

1.  $R_2 \leftarrow R_2 - R_1 \rightarrow R_2 = (0, 1, 1).$  Matrix  $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$

2.  $R_3 \leftarrow R_3 - R_2 \rightarrow R_3 = (0, 0, 0).$  REF:  $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$

3. Two nonzero rows  $\rightarrow$  **rank = 2.**

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### Problem 12 (R.K. Jain & Iyengar)

Matrix:

$$N = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 6 & 9 \\ 2 & 5 & 8 \end{pmatrix}$$

**Task:** REF and rank.

**Solution:**

1.  $R_2 \leftarrow R_2 - 3R_1 \rightarrow R_2 = (0, 0, 0).$

$R_3 \leftarrow R_3 - 2R_1 \rightarrow R_3 = (0, 1, 2).$  Matrix  $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 2 \end{pmatrix}.$

2. Swap  $R_2$  and  $R_3$ :  $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$  REF reached.

3. Two nonzero rows  $\rightarrow$  **rank** = 2.

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### Problem 13 (R.K. Jain & Iyengar)

**Matrix:**

$$P = \begin{pmatrix} 2 & 4 & -1 \\ 1 & 3 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

**Task:** REF and rank.

**Solution:**

1. Swap rows to use  $R_2$  as first row for simpler pivot: swap  $R_1 \leftrightarrow R_2$ :  $\begin{pmatrix} 1 & 3 & 0 \\ 2 & 4 & -1 \\ 0 & 1 & 1 \end{pmatrix}.$

2.  $R_2 \leftarrow R_2 - 2R_1$ :  $R_2 = (0, -2, -1).$  Matrix  $\begin{pmatrix} 1 & 3 & 0 \\ 0 & -2 & -1 \\ 0 & 1 & 1 \end{pmatrix}.$

3.  $R_3 \leftarrow R_3 + \frac{1}{2}R_2$  is an option, but easier:  $R_3 \leftarrow R_3 + \frac{1}{2}R_2 \rightarrow$  compute  $(0, 1, 1) + \frac{1}{2}(0, -2, -1) = (0, 1 - 1, 1 - \frac{1}{2}) = (0, 0, \frac{1}{2}).$  To avoid fractions, instead do  $R_2 \leftarrow -R_2$  to get positive pivot:  $R_2 = (0, 2, 1).$  Then  $R_3 \leftarrow R_3 - (0, 1, 1)$  with multiplier 1 gives  $R_3 - (0, 1, 1) = (0, 0, 0).$  But carefully:

- From step 2 we had  $R_2 = (0, -2, -1).$  Replace  $R_2 \leftarrow -R_2 = (0, 2, 1).$

- $R_3 \leftarrow R_3 - \frac{1}{2}R_2$ :  $R_3 = (0, 1, 1) - \frac{1}{2}(0, 2, 1) = (0, 0, \frac{1}{2}).$

We can instead do  $R_3 \leftarrow 2R_3 - R_2$  to avoid fractions:  $2R_3 = (0, 2, 2),$  subtract  $R_2 = (0, 2, 1) \rightarrow (0, 0, 1).$

So do:

- Multiply  $R_3$  by 2: temporary  $R'_3 = (0, 2, 2).$

- $R'_3 \leftarrow R'_3 - R_2 = (0, 0, 1).$  Replace  $R_3$  by that result and divide by 1:  $R_3 = (0, 0, 1).$

Final REF approximately  $\begin{pmatrix} 1 & 3 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$

4. Three nonzero rows  $\rightarrow$  **rank** = 3.

(Note: multiple valid sequences of row ops exist; final REF and rank same.)

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### Problem 14 (R.K. Jain & Iyengar)

Matrix:

$$Q = \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 2 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

Task: REF and rank.

Solution:

1.  $R_3 \leftarrow R_3 - R_1 - R_2$  (since  $R_1 + R_2 = (1, 1, 1, 1)$ ):  $R_3 = (1, 1, 1, 1) - (1, 1, 1, 1) = (0, 0, 0, 0)$ .

2. Matrix becomes  $\begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ . REF reached.

3. Two nonzero rows  $\rightarrow$  **rank** = 2.

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### Problem 15 (R.K. Jain & Iyengar)

Matrix:

$$R = \begin{pmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 5 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

Task: REF and rank.

Solution:

1.  $R_2 \leftarrow R_2 - 2R_1$ :  $R_2 = (0, 0, -1, 1)$ .

$R_3 \leftarrow R_3 - R_1$ :  $R_3 = (0, -1, -2, 1)$ . Matrix  $\begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & -1 & -2 & 1 \end{pmatrix}$ .

2. Swap  $R_2$  and  $R_3$ :  $\begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & -1 & -2 & 1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$ .

3.  $R_2 \leftarrow -R_2 = (0, 1, 2, -1)$ . Then eliminate below (already zero). REF:  $\begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$ .

4. Three nonzero rows  $\rightarrow$  **rank** = 3.

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### Problem 16 (R.K. Jain & lyengar)

Matrix:

$$S = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 5 & 7 \\ 0 & 1 & 2 & 3 \end{pmatrix}$$

Task: REF and rank.

Solution:

1.  $R_2 \leftarrow R_2 - R_1 \rightarrow R_2 = (0, 1, 2, 3).$

$R_3 \leftarrow R_3 - 2R_1 \rightarrow R_3 = (0, 1, 3, 5).$

$R_4$  is already  $(0, 1, 2, 3).$  Matrix:

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 3 & 5 \\ 0 & 1 & 2 & 3 \end{pmatrix}.$$

2.  $R_3 \leftarrow R_3 - R_2 \rightarrow R_3 = (0, 0, 1, 2).$

$R_4 \leftarrow R_4 - R_2 \rightarrow R_4 = (0, 0, 0, 0).$  Matrix  $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$

3. Three nonzero rows  $\rightarrow$  rank = 3.

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### Problem 17 (R.K. Jain & lyengar)

Matrix:

$$T = \begin{pmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 0 & 1 & 2 \\ 3 & 6 & 9 \end{pmatrix}$$

Task: REF and rank.

Solution:

1. Swap  $R_1 \leftrightarrow R_2$  to use 1 pivot:  $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 2 \\ 3 & 6 & 9 \end{pmatrix}.$

$$2. R_2 \leftarrow R_2 - 2R_1 \rightarrow R_2 = (0, 0, 0).$$

$$R_4 \leftarrow R_4 - 3R_1 \rightarrow R_4 = (0, 0, 0). \text{ Matrix } \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

$$3. \text{ Swap } R_2 \leftrightarrow R_3: \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \text{ REF.}$$

$$4. \text{ Two nonzero rows} \rightarrow \text{rank} = 2.$$


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### Problem 18 (R.K. Jain & Iyengar)

Matrix:

$$U = \begin{pmatrix} 1 & -1 & 2 & 0 \\ 2 & -2 & 4 & 0 \\ 1 & 0 & 3 & 1 \end{pmatrix}$$

Task: REF and rank.

Solution:

$$1. R_2 \leftarrow R_2 - 2R_1 \rightarrow R_2 = (0, 0, 0, 0).$$

$$R_3 \leftarrow R_3 - R_1 \rightarrow R_3 = (0, 1, 1, 1). \text{ Matrix } \begin{pmatrix} 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix}.$$

$$2. \text{ Swap } R_2 \text{ and } R_3: \begin{pmatrix} 1 & -1 & 2 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \text{ REF.}$$

$$3. \text{ Two nonzero rows} \rightarrow \text{rank} = 2.$$


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### Problem 19 (R.K. Jain & Iyengar)

Matrix:

$$V = \begin{pmatrix} 1 & 3 & 1 \\ 2 & 6 & 2 \\ 1 & 4 & 2 \end{pmatrix}$$

Task: REF and rank.

Solution:

$$1. R_2 \leftarrow R_2 - 2R_1 \rightarrow R_2 = (0, 0, 0).$$

$$R_3 \leftarrow R_3 - R_1 \rightarrow R_3 = (0, 1, 1). \text{ Matrix } \begin{pmatrix} 1 & 3 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

$$2. \text{ Swap } R_2 \text{ and } R_3 \rightarrow \begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}. \text{ REF.}$$

$$3. \text{ Two nonzero rows} \rightarrow \text{rank} = 2.$$

## Problem 20 (R.K. Jain & Iyengar)

Matrix:

$$W = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 1 & 5 & 5 \end{pmatrix}$$

Task: REF and rank.

Solution:

$$1. \text{ Use } R_3 \leftarrow R_3 - R_1 - R_2 \text{ because } R_1 + R_2 = (1, 1, 1, 3):$$

$$R_3 = (1, 1, 1, 3) - (1, 1, 1, 3) = (0, 0, 0, 0).$$

$$2. R_4 \leftarrow R_4 - 2R_1: R_4 = (2, 1, 5, 5) - 2(1, 0, 2, 1) = (0, 1, 1, 3). \text{ After the first step matrix}$$

$$\text{becomes } \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 3 \end{pmatrix}.$$

$$3. R_4 \leftarrow R_4 - R_2 \rightarrow R_4 = (0, 0, 2, 1). \text{ So now } \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 \end{pmatrix}.$$

$$4. \text{ Swap } R_3 \text{ and } R_4 \text{ to move nonzero row up: } \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \text{ — REF.}$$

$$5. \text{ Three nonzero rows} \rightarrow \text{rank} = 3.$$

## Notes on sources and usage

- I labeled each problem with the requested textbook attribution beside the problem heading. The problems were chosen and written *in the style of* typical exercise problems found in those textbooks (Grewal; Jain & Iyengar).

- Many matrices admit multiple valid sequences of elementary row operations; I presented a clean, arithmetic-friendly path in each case (avoiding fractions where possible, showing equivalent integer-row manipulations).
- For each matrix I stopped at Row Echelon Form (REF) — i.e., zeros below pivots — since that is the requested form for rank determination. If you want **reduced row echelon form (RREF)** for any selected problems I can produce that as well.